

Runda Liu

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EDUCATION

Xidian University, China

B.Eng in Electronic and Information Engineering | GPA 3.8/4.0

09/2022– 06/2026

Relevant Coursework: Digital Circuit | RF Circuit | Data Structure | Python | Computer Vision

Scholarly Communication: Participated a Scholarly Program at Massachusetts Institute of Technology, USA

Outstanding Student of ECE-Class of 2026 in NUS (Suzhou) Research Institute.

INTERNSHIP

Sentigent Tech, Computer Vision Intern – Jiangsu Suzhou, China

04/2026– 07/2026

- **Optimize RTMW for accurate 2D/3D keypoints regression.** Constructed a 200M paired 2D & 3D human keypoints dataset. The RTMW detector outputs 48 upper-body 2D landmarks (with hands) and 18 full-body 3D landmarks, reaching **46 mm Masked-MPJPE (3D)** and **92% PCK@5% (2D)**.
- **Monocular Depth Distillation.** Distilled MobileNetV2 using DepthAnythingV3 (DA3) teacher model on TartanGround (100M samples) for relative depth prediction via a depth-ray dual-head decoder. Without sky pixels, the student model hits **$\delta_1=0.6$ (87.7% of DA3)** and **96% feature SSIM**.
- **Build multiple business-related virtual scenes on UE.** Implement customized depth pass processing for translucent materials such as water surfaces and glass on **Unreal Engine**. Leverage **AirSim** to record trajectory metadata and collect stereo RGB-depth data from simulated vehicles.

PROJECTS

Deep reinforcement learning algorithm for multi-platform (UAVs, Radars, etc) detection and attack planning

Xi'an, China

10/2024 – 07/2025

- Build reinforcement learning simulation environment and design multi-objective reward function (benefit, platform load balance, distance penalty, etc.).
- For the task allocation problem of the platform, each platform is modeled as an independent agent, and an independent actor network is designed for it.
- The training is based on **MAPPO (Multi-Agent Proximal Policy Optimization)** algorithm, and the action mask mechanism is utilized to restrict illegal actions, which improves the ability of strategy learning.
- Compared with the doctoral thesis algorithm, the collaborative task planning performance of the algorithm is improved by about **11.26%**.

Near/Far Field Antenna Array Based on MATLAB Design of Computational Function Modules

Xi'an, China

03/2024 – 03/2025

- Built mathematical models for antenna arrays and derived electromagnetic computational algorithms leveraging far-field pattern multiplication and near-field field superposition theorems.
- Implement an algorithm to calculate the near-field electromagnetic energy distribution on a specified distance observation plane and generate a visual hot spot map.
- Implement the algorithm to calculate the far-field three-dimensional radiation pattern (PAT), support the decomposition of array factors (AF) and element factors (EP), and visualize them in sine space.

AWARDS

- **2022-2023 school year third class scholarship**
- **The third prize of Shaanxi Province in the 2024 National Undergraduate Mathematics Competition**

SKILLS

Programming: Python, Matlab.

Framework & Tools: Linux, PyTorch, TensorFlow, Cuda, ComputerVision, Blender, Unreal Engine.